

Hessian Geometry, Local Rates, and Acceleration of Sinkhorn

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Abstract

This note isolates one conditional-expectation operator that controls three aspects of entropic optimal transport: dual curvature, the local convergence of Sinkhorn's algorithm, and semi-dual conditioning. At the optimal coupling, the positive full-dual curvature operator is $\epsilon^{-1} \begin{pmatrix} I & T \\ T^* & I \end{pmatrix}$, the Jacobian of one complete Sinkhorn cycle is T^*T , and exact elimination of one potential gives the semi-dual Hessian $\epsilon^{-1}(I - T^*T)$. This common structure yields the local rate, strict-gap criteria, small- ϵ asymptotics, and explicit parameters for extrapolation and momentum. It also explains why variable projection and quasi-Newton methods are natural alternatives to long scaling iterations.

1 Dual Hessian and conditional expectation

The correct geometry for the dual potentials is not an unweighted Euclidean space: it is the product of the two marginal L^2 spaces. In this geometry, the Hessian has an especially simple block form.

Entropic primal and dual. Let $\mathcal{X}$ and $\mathcal{Y}$ be standard Borel spaces, let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and let $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{+\infty\}$ be measurable. The KL-regularized transport value is

$$\mathcal{L}_c^\epsilon(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \left\{ \int c \, d\pi + \epsilon \, \text{KL}(\pi \parallel \alpha \otimes \beta) \right\}. \quad (1)$$

This is the entropy-regularized transport problem introduced computationally by Cuturi [3]. Its concave dual objective is

$$\mathcal{D}_\epsilon(f, g) := \int_{\mathcal{X}} f \, d\alpha + \int_{\mathcal{Y}} g \, d\beta - \epsilon \int_{\mathcal{X} \times \mathcal{Y}} \left[\exp \left(\frac{f(x) + g(y) - c(x, y)}{\epsilon} \right) - 1 \right] d\alpha(x) d\beta(y) \quad (2)$$

and $\mathcal{L}_c^\epsilon(\alpha, \beta) = \sup_{f, g} \mathcal{D}_\epsilon(f, g)$ under the standard duality assumptions. Assume henceforth that an optimal pair (f^*, g^*) exists and that its Gibbs coupling

$$d\pi^*(x, y) = \exp \left(\frac{f^*(x) + g^*(y) - c(x, y)}{\epsilon} \right) d\alpha(x) d\beta(y) \quad (3)$$

has marginals (α, β) . The gauge transformation $(f, g) \mapsto (f + s, g - s)$ leaves (2) unchanged. We isolate the nonconstant correlation modes through the centered spaces

$$L_0^2(\alpha) := \left\{ h : \int h \, d\alpha = 0 \right\}, \quad L_0^2(\beta) := \left\{ k : \int k \, d\beta = 0 \right\},$$

endowed with their usual weighted norms. We write

$$\text{cent}_\alpha h := h - \int h \, d\alpha, \quad \text{cent}_\beta k := k - \int k \, d\beta \quad (4)$$

for the orthogonal centering projectors. The full quotient by the gauge also retains the common-constant direction $(\mathbf{1}, \mathbf{1})$; its curvature will be recorded separately.

Conditional expectation. Disintegrating $\pi^\star$ defines

$$T : L^2(\beta) \longrightarrow L^2(\alpha), \quad Tk(x) := \mathbb{E}_{\pi^\star}[k(Y) \mid X = x], \quad (5)$$

$$T^\star : L^2(\alpha) \longrightarrow L^2(\beta), \quad T^\star h(y) := \mathbb{E}_{\pi^\star}[h(X) \mid Y = y]. \quad (6)$$

The notation is literal: $\langle h, Tk \rangle_\alpha = \langle T^\star h, k \rangle_\beta$. Conditional Jensen gives $\|T\|_{\text{op}} \leq 1$, while $T\mathbf{1} = \mathbf{1}$ and $T^\star\mathbf{1} = \mathbf{1}$. Hence both operators preserve centered functions. We use the same symbols for their restrictions and set

$$\sigma_\epsilon := \left\| T : L_0^2(\beta) \rightarrow L_0^2(\alpha) \right\|_{\text{op}}, \quad \gamma_\epsilon := 1 - \sigma_\epsilon^2. \quad (7)$$

Thus σ_ϵ is the maximal nonconstant correlation carried by $\pi^\star$, while γ_ϵ is the spectral gap of the Markov operator $T^\star T$ on $L_0^2(\beta)$.

Proposition 1.1 (Full dual Hessian). *In $L^2(\alpha) \oplus L^2(\beta)$, the positive curvature operator of the concave dual at $(f^\star, g^\star)$ is*

$$H_{\text{dual}} = -\nabla^2 \mathcal{D}_\epsilon(f^\star, g^\star) = \frac{1}{\epsilon} \begin{pmatrix} I & T \\ T^\star & I \end{pmatrix}. \quad (8)$$

Equivalently,

$$-\epsilon D^2 \mathcal{D}_\epsilon(f^\star, g^\star)[(h, k), (h, k)] = \mathbb{E}_{\pi^\star}[(h(X) + k(Y))^2]. \quad (9)$$

On $L_0^2(\alpha) \oplus L_0^2(\beta)$,

$$\frac{1 - \sigma_\epsilon}{\epsilon} I \preceq H_{\text{dual}} \preceq \frac{1 + \sigma_\epsilon}{\epsilon} I, \quad \kappa_{\text{dual},0} = \frac{1 + \sigma_\epsilon}{1 - \sigma_\epsilon}. \quad (10)$$

On the full space modulo the gauge direction $(\mathbf{1}, -\mathbf{1})$, the additional eigenvalue is $2/\epsilon$, and hence

$$\kappa_{\text{dual}} = \frac{2}{1 - \sigma_\epsilon}. \quad (11)$$

Proof. Twice differentiating $-\mathcal{D}_\epsilon$ gives $\epsilon^{-1} \mathbb{E}_{\pi^\star}[(h(X) + k(Y))^2]$. Expanding the square and using (5) gives (8). On a singular pair $Tv = su$, $T^\star u = sv$, the two vectors $(u, \pm v)$ have eigenvalues $(1 \pm s)/\epsilon$. Taking the extremal centered singular value $s = \sigma_\epsilon$ proves (10). The omitted constant gauge vector $(\mathbf{1}, -\mathbf{1})$ has eigenvalue zero, whereas $(\mathbf{1}, \mathbf{1})$ has eigenvalue $2/\epsilon$. This proves (11). $\square$

For discrete marginals $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, with optimal coupling matrix $P^\star$, the same operators read

$$Tk = D_a^{-1} P^\star k, \quad T^\star h = D_b^{-1} (P^\star)^\top h,$$

where $D_a = \text{diag}(a)$ and $D_b = \text{diag}(b)$. These formulas are adjoint in the a - and b -weighted inner products, not in the unweighted Euclidean ones.

2 Sinkhorn Jacobian and local rate

A full Sinkhorn cycle is a nonlinear map on one potential. Its linearization at the fixed point is exactly the two-step conditional expectation $T^\star T$, so the local convergence question reduces to a Markov spectral gap.

Soft transforms and the cycle map. Exact block maximization of $\mathcal{D}_\epsilon$ is written with the continuous soft c -transforms

$$g^{\bar{c},\epsilon}(x) := -\epsilon \log \int_{\mathcal{Y}} \exp\left(\frac{g(y) - c(x, y)}{\epsilon}\right) d\beta(y), \quad (12)$$

$$f^{c,\epsilon}(y) := -\epsilon \log \int_{\mathcal{X}} \exp\left(\frac{f(x) - c(x, y)}{\epsilon}\right) d\alpha(x). \quad (13)$$

One complete cycle on the target potential is therefore

$$\mathcal{S}_\epsilon(g) := (g^{\bar{c},\epsilon})^{c,\epsilon}, \quad g^{(\ell+1)} = \mathcal{S}_\epsilon(g^{(\ell)}). \quad (14)$$

The opposite ordering acts on $L^2(\alpha)$ and has the same nonzero spectrum.

Proposition 2.1 (Jacobian and exact local Sinkhorn rate). *Assume that the two soft transforms are twice differentiable near the optimum. Then*

$$D[g \mapsto g^{\bar{c},\epsilon}](g^\star) = -T, \quad D[f \mapsto f^{c,\epsilon}](f^\star) = -T^*, \quad D\mathcal{S}_\epsilon(g^\star) = T^*T. \quad (15)$$

Consequently, for the centered error $e^{(\ell)} = \text{cent}_\beta(g^{(\ell)} - g^\star)$ of any Sinkhorn sequence converging to $g^\star$ modulo constants,

$$e^{(\ell+1)} = T^*T e^{(\ell)} + o(\|e^{(\ell)}\|_\beta), \quad \limsup_{\ell \rightarrow \infty} \frac{\|e^{(\ell+1)}\|_\beta}{\|e^{(\ell)}\|_\beta} \leq \sigma_\epsilon^2. \quad (16)$$

If σ_ϵ^2 is an isolated eigenvalue and the asymptotic error has a nonzero projection onto its eigenspace, the ratio converges to σ_ϵ^2 . Moreover,

$$\gamma_\epsilon = \inf_{\substack{k \in L_0^2(\beta) \\ \|k\|_\beta=1}} \mathbb{E}_{\pi^\star} [\text{Var}(k(Y) \mid X)]. \quad (17)$$

Proof. Differentiating the log-partition maps (12)–(13) at the optimal Gibbs coupling gives the first two identities in (15); the chain rule gives the third. Taylor expansion proves (16). Finally, for centered k ,

$$\langle k, (I - T^*T)k \rangle_\beta = \|k\|_\beta^2 - \|Tk\|_\alpha^2 = \mathbb{E}_{\pi^\star} [\text{Var}(k(Y) \mid X)].$$

Taking the infimum over the unit sphere gives (17). $\square$

The reverse cycle $f \mapsto (f^{c,\epsilon})^{\bar{c},\epsilon}$ has Jacobian TT^* on $L_0^2(\alpha)$. Since T^*T and TT^* have the same nonzero spectrum, either orientation has the same local rate. In finite spaces, this recovers the classical second-singular-value formula for matrix scaling [12, 13].

When is the rate strictly smaller than one? Positivity of the Gibbs kernel is enough in finite dimension but not on an arbitrary infinite space: centered approximate modes may retain correlation arbitrarily close to one. The following input-level bound is simple and explicit.

Define the cost oscillation modulo separable terms by

$$\Omega_c := \inf_{\xi, \zeta} \text{ess osc}_{\alpha \otimes \beta} (c(x, y) - \xi(x) - \zeta(y)). \quad (18)$$

Here the infimum is over measurable real-valued functions ξ on $\mathcal{X}$ and ζ on $\mathcal{Y}$. For compact supports and continuous c , also define the invariant four-point oscillation

$$\Delta_c := \sup_{x, x', y, y'} [c(x, y') + c(x', y) - c(x, y) - c(x', y')], \quad (19)$$

where the supremum is over the product of the two supports.

Proposition 2.2 (Explicit strict-gap criteria). *If $\Omega_c < \infty$, then*

$$\sigma_\epsilon \leq 1 - e^{-2\Omega_c/\epsilon} < 1, \quad \sigma_\epsilon^2 \leq (1 - e^{-2\Omega_c/\epsilon})^2. \quad (20)$$

If the supports are compact and c is continuous, then the projective bound also holds:

$$\sigma_\epsilon \leq \Lambda_\epsilon := \tanh\left(\frac{\Delta_c}{4\epsilon}\right) < 1. \quad (21)$$

Proof. Adding $\xi(x) + \zeta(y)$ to c only changes the dual potentials. For any $\eta > 0$, choose such terms so that the resulting cost has essential oscillation at most $\Omega_c + \eta$. Write $k_\epsilon = e^{-c/\epsilon}$, $m = \text{ess inf } k_\epsilon$, $M = \text{ess sup } k_\epsilon$, and $d\pi^* = u(x)k_\epsilon(x, y)v(y) d\alpha d\beta$. The two marginal equations give

$$\frac{1}{MV} \leq u \leq \frac{1}{mV}, \quad \frac{1}{MU} \leq v \leq \frac{1}{mU}, \quad \frac{1}{M} \leq UV \leq \frac{1}{m},$$

where $U = \int u d\alpha$ and $V = \int v d\beta$. Hence the density $r = d\pi^*/d(\alpha \otimes \beta)$ satisfies $r \geq (m/M)^2 \geq e^{-2(\Omega_c + \eta)/\epsilon} =: \delta_\eta$. Decomposing $r = \delta_\eta + (1 - \delta_\eta)\tilde{r}$ shows that, on centered functions, $T = (1 - \delta_\eta)\tilde{T}$ with $\|\tilde{T}\|_{\text{op}} \leq 1$. Letting $\eta \downarrow 0$ proves (20).

Under compactness, the conditional kernel is continuous and T is compact. Birkhoff's theorem bounds both half-cycles in Hilbert's projective metric by Λ_ϵ [7]. If $\sigma_\epsilon > 0$, compactness provides a continuous centered singular pair $Tv = \sigma_\epsilon u$, $T^*u = \sigma_\epsilon v$. Differentiating the two projective bounds gives $\sigma_\epsilon \text{osc}(u) \leq \Lambda_\epsilon \text{osc}(v)$ and the reverse inequality with u, v exchanged. Multiplication yields $\sigma_\epsilon^2 \leq \Lambda_\epsilon^2$; the case $\sigma_\epsilon = 0$ is immediate. $\square$

On unbounded spaces, cost growth must be balanced against marginal tails. Conforti, Durmus, and Greco prove quantitative contraction for quadratic costs under asymptotically positive log-concavity profiles [2]; Eckstein obtains weighted Hilbert contraction under compatible cost-growth and tail assumptions [6]. These are useful input-level criteria, but their norms should not be identified automatically with the exact L^2 gap (17).

Why a linear small- ϵ gap is natural. In a smooth connected Monge regime, the conditional law of Y given $X = x$ has width $O(\sqrt{\epsilon})$ around the hard transport map. The conditional variance in (17) is therefore typically of order ϵ , leading to $\sigma_\epsilon^2 = 1 - C\epsilon + o(\epsilon)$.

A precise conditional statement is as follows. Assume that $\mathcal{X}$ and $\mathcal{Y}$ are smooth connected d -dimensional manifolds, that the marginals have positive smooth densities, and that hard transport is induced by a smooth diffeomorphism $M : \mathcal{X} \rightarrow \mathcal{Y}$. Let g_0^* be the hard target potential and set

$$H(x) := D_{yy}^2(c(x, \cdot) - g_0^*)(M(x)), \quad A(y) := H(M^{-1}(y))^{-1}. \quad (22)$$

Suppose that H is uniformly positive and that the uniform Laplace expansion of the entropic conditionals permits passage to the first nonzero Rayleigh quotient, for instance through Mosco convergence of the rescaled forms and compact convergence of their resolvents. Define

$$\lambda_1(A, \beta) := \inf_{\int k d\beta=0} \int_{\mathcal{Y}} \langle A(y) \nabla k(y), \nabla k(y) \rangle d\beta(y). \quad (23)$$

Uniform ellipticity and a Poincaré inequality for β ensure that $\lambda_1(A, \beta) > 0$.

Proposition 2.3 (Smooth small-regularization law). *Under the preceding assumptions,*

$$\gamma_\epsilon = \epsilon \lambda_1(A, \beta) + o(\epsilon), \quad \sigma_\epsilon^2 = 1 - \epsilon \lambda_1(A, \beta) + o(\epsilon), \quad \sigma_\epsilon = 1 - \frac{\epsilon}{2} \lambda_1(A, \beta) + o(\epsilon). \quad (24)$$

Proof. Writing π_ϵ^* for the optimal coupling at regularization ϵ , the uniform Laplace expansion of $Y \mid X = x$ gives $\text{Cov}(Y \mid X = x) = \epsilon H(x)^{-1} + o(\epsilon)$. Hence, for smooth k ,

$$\frac{1}{\epsilon} \mathbb{E}_{\pi_\epsilon^*}[\text{Var}(k(Y) \mid X)] \longrightarrow \int_{\mathcal{X}} \left\langle H(x)^{-1} \nabla k(\mathbf{M}(x)), \nabla k(\mathbf{M}(x)) \right\rangle d\alpha(x).$$

Since $\mathbf{M}_\# \alpha = \beta$, the right-hand side is the form in (23). Spectral convergence permits passage to the first Rayleigh quotient in (17), giving the first expansion. The other two follow from $\gamma_\epsilon = 1 - \sigma_\epsilon^2$. $\square$

Example 2.4 (Exact Gaussian benchmark). For $\alpha = \beta = \mathcal{N}(0, 1)$ and $c(x, y) = \frac{1}{2}(x - y)^2$, the optimal coupling is Gaussian with correlation

$$r_\epsilon = \frac{\sqrt{\epsilon^2 + 4} - \epsilon}{2}. \quad (25)$$

Hermite polynomials diagonalize T , so $\sigma_\epsilon = r_\epsilon$. Thus $\sigma_\epsilon = 1 - \epsilon/2 + O(\epsilon^2)$ and $\sigma_\epsilon^2 = 1 - \epsilon + O(\epsilon^2)$. This is Proposition 2.3 with $A = 1$ and $\lambda_1(A, \beta) = 1$.

Local L^2 versus global Hilbert contraction. For $\Delta_c > 0$,

$$1 - \Lambda_\epsilon^2 \sim 4e^{-\Delta_c/(2\epsilon)} \quad (\epsilon \downarrow 0). \quad (26)$$

This is exponentially more pessimistic than the smooth local law $\gamma_\epsilon \sim \epsilon \lambda_1(A, \beta)$. There is no contradiction. Hilbert's projective metric gives a global guarantee over the full positive cone and uses no smoothness of the iterates. The L^2 result is local to the optimal coupling and exploits the regular diffusive structure of the dominant modes. Chizat, Delalande, and Vaškevičius similarly obtain polynomial rather than universal exponential dependences under continuous regularity and log-concavity assumptions [1].

3 Extrapolation and momentum

Once the local spectrum is reduced to $\text{spec}(T^*T) \subset [0, \sigma_\epsilon^2]$, standard polynomial acceleration applies. This section distinguishes scalar extrapolation, which improves the gap by a constant factor, from momentum, which gives the characteristic square-root improvement.

Scalar extrapolation. Consider the relaxed full cycle

$$g^{(\ell+1)} = g^{(\ell)} + \omega(\mathcal{S}_\epsilon(g^{(\ell)}) - g^{(\ell)}). \quad (27)$$

Its centered Jacobian is $I - \omega(I - T^*T)$.

Proposition 3.1 (Optimal scalar extrapolation). *If $\gamma_\epsilon > 0$, the choice*

$$\omega_{\text{ext}} = \frac{2}{1 + \gamma_\epsilon} \quad (28)$$

minimizes the worst local factor of (27) over the spectral interval $[0, 1 - \gamma_\epsilon]$. The resulting factor is

$$r_{\text{ext}} = \frac{1 - \gamma_\epsilon}{1 + \gamma_\epsilon} = \frac{\sigma_\epsilon^2}{2 - \sigma_\epsilon^2}, \quad (29)$$

compared with $\sigma_\epsilon^2 = 1 - \gamma_\epsilon$ for ordinary Sinkhorn.

Proof. On an eigenmode $T^*Tk = \lambda k$, the factor is $1 - \omega(1 - \lambda)$, with $1 - \lambda \in [\gamma_\epsilon, 1]$. The minimax choice balances the two endpoint values: $1 - \omega\gamma_\epsilon = -(1 - \omega)$. Solving gives (28) and (29). $\square$

For $\gamma_\epsilon \ll 1$, scalar extrapolation changes $1 - \gamma_\epsilon$ into $1 - 2\gamma_\epsilon + O(\gamma_\epsilon^2)$. It is useful but does not change the scaling of the iteration count.

Momentum and square-root acceleration. Add one memory term to the extrapolated cycle:

$$g^{(\ell+1)} = g^{(\ell)} + \omega(\mathcal{S}_\epsilon(g^{(\ell)}) - g^{(\ell)}) + \vartheta(g^{(\ell)} - g^{(\ell-1)}). \quad (30)$$

This is the heavy-ball polynomial applied locally to $I - T^*T$ [16].

Proposition 3.2 (Optimal local momentum parameters). *For a known gap $\gamma_\epsilon > 0$, choose*

$$\omega_{\text{mom}} = \frac{4}{(1 + \sqrt{\gamma_\epsilon})^2}, \quad \vartheta_{\text{mom}} = \left(\frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}} \right)^2. \quad (31)$$

Then the local asymptotic factor of (30) is

$$r_{\text{mom}} = \frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}}. \quad (32)$$

Thus the convergence gap improves from order γ_ϵ to order $\sqrt{\gamma_\epsilon}$.

Proof. For a mode with $a = 1 - \lambda \in [\gamma_\epsilon, 1]$, the scalar recurrence is

$$z_{\ell+1} = (1 - \omega a + \vartheta)z_\ell - \vartheta z_{\ell-1}.$$

With (31), put $r = (1 - \sqrt{\gamma_\epsilon})/(1 + \sqrt{\gamma_\epsilon})$. Then $\vartheta = r^2$ and $1 - \omega a + \vartheta$ ranges from $2r$ to $-2r$ as a ranges from γ_ϵ to 1. Both characteristic roots therefore have modulus at most r . $\square$

Over-relaxing the two block updates separately gives the nonlinear SOR variant studied in [18, 13]. Its optimal local parameter is

$$\omega_{\text{SOR}} = \frac{2}{1 + \sqrt{\gamma_\epsilon}},$$

and it has the same factor (32). Block SOR and the outer momentum step (30) are different algorithms, despite their identical optimal local rate. Their global convergence requires safeguards; the displayed parameters are local spectral prescriptions, not unconditional global guarantees.

Estimating the gap from iterates. The optimal parameters depend on an unknown local quantity. During an ordinary Sinkhorn run, set

$$d^{(\ell)} := \text{cent}_\beta(g^{(\ell)} - g^{(\ell-1)}).$$

Near the fixed point, $d^{(\ell+1)} = T^*T d^{(\ell)} + o(\|d^{(\ell)}\|)$, so the Rayleigh quotient

$$\hat{r}_\ell := \frac{\langle d^{(\ell)}, d^{(\ell+1)} \rangle_\beta}{\|d^{(\ell)}\|_\beta^2} \quad (33)$$

converges generically to σ_ϵ^2 when a dominant mode is present. The norm ratio $\|d^{(\ell+1)}\|_\beta / \|d^{(\ell)}\|_\beta$ is a simpler, slightly noisier alternative. In practice one clips $\hat{r}_\ell$ to $[0, 1]$, sets $\hat{\gamma}_\ell = 1 - \hat{r}_\ell$, and activates acceleration only after the estimate has stabilized. This also explains why extrapolation based on early iterates can be unreliable: the asymptotic spectral mode has not yet emerged.

4 Semi-dual variable projection

Eliminating one potential exactly converts the same operator T^*T from an iteration Jacobian into a Hessian correction. This is the variable-projection mechanism: the poorly conditioned block coupling is removed by a Schur complement rather than handled by alternating block maximization.

Reduced objective and Hessian. For each g , the exact source-potential maximizer is $g^{\bar{c}, \epsilon}$ from (12). The resulting concave semi-dual is

$$\mathcal{E}_\epsilon(g) := \mathcal{D}_\epsilon(g^{\bar{c}, \epsilon}, g) = \int_{\mathcal{X}} g^{\bar{c}, \epsilon} d\alpha + \int_{\mathcal{Y}} g d\beta. \quad (34)$$

Let

$$d\pi_g(x, y) := \exp\left(\frac{g^{\bar{c}, \epsilon}(x) + g(y) - c(x, y)}{\epsilon}\right) d\alpha(x) d\beta(y), \quad (35)$$

and denote its second marginal by β_g . Its first marginal is exactly α .

Proposition 4.1 (Semi-dual derivatives and Schur complement). *For bounded directions k, k' on $\mathcal{Y}$,*

$$D\mathcal{E}_\epsilon(g)[k] = \int_{\mathcal{Y}} k d(\beta - \beta_g), \quad (36)$$

$$D^2\mathcal{E}_\epsilon(g)[k, k'] = -\frac{1}{\epsilon} \int_{\mathcal{X}} \text{Cov}_{\pi_g(\cdot|x)}(k(Y), k'(Y)) d\alpha(x). \quad (37)$$

At $g = g^*$, the positive Riesz curvature operator on $L^2(\beta)$ is

$$H_{\text{sd}} := -\nabla^2 \mathcal{E}_\epsilon(g^*) = \frac{1}{\epsilon} (I - T^*T). \quad (38)$$

The identity extends from bounded directions to $L^2(\beta)$ by density. It is exactly the Schur complement of the f block in (8).

Proof. The envelope theorem gives (36). Differentiating the conditional Gibbs law in (35) gives the conditional covariance formula (37). At the optimum,

$$\int_{\mathcal{X}} \text{Cov}(k, k' | X) d\alpha = \langle k, k' \rangle_\beta - \langle Tk, Tk' \rangle_\alpha = \langle k, (I - T^*T)k' \rangle_\beta,$$

which proves (38). Finally, the Schur complement of I in $\begin{pmatrix} I & T \\ T^* & I \end{pmatrix}$ is $I - T^*T$. This is exact variable projection in the sense of Golub and Pereyra [8, 9]. $\square$

Conditioning and first-order rates. The spectral gap that controls Sinkhorn is exactly the strong concavity of $\mathcal{E}_\epsilon$, or equivalently the strong convexity of $-\mathcal{E}_\epsilon$.

Proposition 4.2 (Semi-dual conditioning and optimization rates). *On $L_0^2(\beta)$,*

$$\frac{\gamma_\epsilon}{\epsilon} I \preceq H_{\text{sd}} \preceq \frac{1}{\epsilon} I, \quad \kappa_{\text{sd}} \leq \frac{1}{\gamma_\epsilon}. \quad (39)$$

Gradient ascent on the local quadratic model of $\mathcal{E}_\epsilon$, with optimal constant step $2\epsilon/(1 + \gamma_\epsilon)$, has factor

$$r_{\text{GD}} \leq \frac{1 - \gamma_\epsilon}{1 + \gamma_\epsilon}. \quad (40)$$

Heavy-ball or Chebyshev acceleration has factor

$$r_{\text{HB}} \leq \frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}}. \quad (41)$$

Proof. The spectrum of T^*T on centered functions lies in $[0, 1 - \gamma_\epsilon]$, so the spectrum of (38) lies in $[\gamma_\epsilon/\epsilon, 1/\epsilon]$. This proves the curvature and condition-number bounds. The standard minimax polynomials on an interval give (40) and (41). $\square$

Since $\gamma_\epsilon = (1 - \sigma_\epsilon)(1 + \sigma_\epsilon)$, this condition-number bound is smaller than the doubly centered full-dual condition number in (10) by the factor $(1 + \sigma_\epsilon)^2$.

The two rates are not merely analogous to those in Section 3: they are the same formulas. Locally,

$$\mathcal{S}_\epsilon(g) - g = \epsilon \nabla \mathcal{E}_\epsilon(g) + o(\|g - g^*\|),$$

where the gradient is the $L^2(\beta)$ Riesz gradient. Thus optimal scalar extrapolation of Sinkhorn is gradient ascent on the semi-dual linearization, and (30) is its heavy-ball counterpart. Nesterov acceleration gives the same square-root dependence on γ_ϵ [15], although its standard worst-case constant differs from the stationary heavy-ball factor (41).

Quasi-Newton methods. In a finite discretization, $-\mathcal{E}_\epsilon$ is a smooth unconstrained convex problem after fixing its additive gauge. BFGS attempts to learn the inverse of $I - T^*T$ from gradient differences, precisely the operator responsible for the slow modes. Under the standard local smoothness and positive-definiteness assumptions, full BFGS is locally superlinear; limited-memory BFGS trades that guarantee for lower memory and often strong practical performance [14].

This strategy has a substantial OT history. Cuturi and Peyré used smooth dual and semi-dual formulations with L-BFGS for variational Wasserstein problems and developed the general regularized semi-dual viewpoint [4, 5]; stochastic first-order optimization of semi-duals was developed by Genevay et al. [10]. Khamlich, Romor, and Rozza use L-BFGS as the optimization core of a modern entropy-regularized semi-discrete solver, together with truncation, multilevel continuation, and fast geometric queries [11]. Qiu, Yin, and Wang also report and analyze L-BFGS on the fully discrete entropic dual as an alternative to long Sinkhorn runs [17].

These methods do not make the small- ϵ conditioning disappear: the inverse Hessian still grows like $1/\gamma_\epsilon$. Their advantage is that curvature information is learned and reused. Each quasi-Newton step is more expensive than one scaling update, so comparisons should count kernel evaluations or matrix-vector products rather than iterations. The semi-dual is especially attractive when one block can be eliminated analytically or integrated accurately, as in semi-discrete transport.

Operator summary. All local statements in the note follow from one contraction:

Object	Operator at the optimum	Consequence
Full dual Hessian	$\epsilon^{-1} \begin{pmatrix} I & T \\ T^* & I \end{pmatrix}$	coupled conditioning
Sinkhorn full-cycle Jacobian	T^*T	local factor σ_ϵ^2
Semi-dual Hessian	$\epsilon^{-1}(I - T^*T)$	curvature γ_ϵ/ϵ
Accelerated local polynomial	$I - T^*T$	rate governed by $\sqrt{\gamma_\epsilon}$

Thus the slow convergence of Sinkhorn and the ill-conditioning of the semi-dual are complementary descriptions of the same near-deterministic correlation modes. Variable projection removes one coupled block; extrapolation changes the first-order polynomial; momentum and quasi-Newton methods approximate the inverse curvature more aggressively.

References

- [1] L. Chizat, A. Delalande, and T. Vaškevičius, Sharper exponential convergence rates for Sinkhorn’s algorithm in continuous settings, *Mathematical Programming*, 215:809–858, 2026. DOI: 10.1007/s10107-025-02242-z.

- [2] G. Conforti, A. Durmus, and G. Greco, Quantitative contraction rates for Sinkhorn’s algorithm: beyond bounded costs and compact marginals, *Annals of Applied Probability*, to appear, 2026. arXiv:2304.04451.
- [3] M. Cuturi, Sinkhorn distances: Lightspeed computation of optimal transport, in *Advances in Neural Information Processing Systems 26*, pages 2292–2300, 2013.
- [4] M. Cuturi and G. Peyré, A smoothed dual approach for variational Wasserstein problems, *SIAM Journal on Imaging Sciences*, 9(1):320–343, 2016. DOI: 10.1137/15M1032600.
- [5] M. Cuturi and G. Peyré, Semidual regularized optimal transport, *SIAM Review*, 60(4):941–965, 2018. DOI: 10.1137/18M1208654.
- [6] S. Eckstein, Hilbert’s projective metric for functions of bounded growth and exponential convergence of Sinkhorn’s algorithm, *Probability Theory and Related Fields*, 193:585–621, 2025. DOI: 10.1007/s00440-025-01366-9.
- [7] J. Franklin and J. Lorenz, On the scaling of multidimensional matrices, *Linear Algebra and its Applications*, 114–115:717–735, 1989.
- [8] G. H. Golub and V. Pereyra, The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate, *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973. DOI: 10.1137/0710036.
- [9] G. H. Golub and V. Pereyra, Separable nonlinear least squares: The variable projection method and its applications, *Inverse Problems*, 19(2):R1–R26, 2003.
- [10] A. Genevay, M. Cuturi, G. Peyré, and F. Bach, Stochastic optimization for large-scale optimal transport, in *Advances in Neural Information Processing Systems 29*, 2016.
- [11] M. Khamlich, F. Romor, and G. Rozza, Efficient numerical strategies for entropy-regularized semi-discrete optimal transport, *Computer Methods in Applied Mechanics and Engineering*, 453:118821, 2026. DOI: 10.1016/j.cma.2026.118821.
- [12] P. A. Knight, The Sinkhorn–Knopp algorithm: Convergence and applications, *SIAM Journal on Matrix Analysis and Applications*, 30(1):261–275, 2008. DOI: 10.1137/060659624.
- [13] T. Lehmann, M.-K. von Renesse, A. Sambale, and A. Uschmajew, A note on overrelaxation in the Sinkhorn algorithm, *Optimization Letters*, 16(8):2209–2220, 2022. DOI: 10.1007/s11590-021-01830-0.
- [14] J. Nocedal and S. J. Wright, *Numerical Optimization*, second edition, Springer, 2006.
- [15] Y. Nesterov, A method for solving the convex programming problem with convergence rate $O(1/k^2)$, *Soviet Mathematics Doklady*, 27:372–376, 1983.
- [16] B. T. Polyak, Some methods of speeding up the convergence of iteration methods, *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964.
- [17] Y. Qiu, H. Yin, and X. Wang, Efficient, stable, and analytic differentiation of the Sinkhorn loss, ICLR submission, OpenReview, 2023. OpenReview: uATOkwOZaI.
- [18] A. Thibault, L. Chizat, C. Dossal, and N. Papadakis, Overrelaxed Sinkhorn–Knopp algorithm for regularized optimal transport, *Algorithms*, 14(5):143, 2021. DOI: 10.3390/a14050143.